

PROJECT REPORT

CHAPTER-1 : INTRODUCTION

1.1 Project Overview

The **India Agricultural Crop Production Analysis (1997–2021)** project is a data analytics and visualization project developed using Tableau. The project focuses on analyzing agricultural crop production trends in India over a period of 25 years, from 1997 to 2021.

The primary objective of this project is to provide comprehensive insights into agricultural production patterns by examining various factors such as:

- State-wise agricultural production
- Crop-wise production trends
- Seasonal crop distribution
- Year-wise production growth
- High and low yielding crops
- Comparative analysis of different agricultural regions

The project utilizes interactive dashboards, calculated fields, filters, KPIs, and storytelling features of Tableau to present meaningful insights. Three dashboards and six story points were developed to enable users to explore agricultural trends efficiently.

The analysis is designed to support various stakeholders, including:

- **Farmers**, to understand crop performance and production patterns.
- **Government policymakers**, to formulate agricultural policies and resource allocation strategies.
- **Researchers and analysts**, to identify trends and conduct further agricultural studies.

The project demonstrates how data analytics and visualization can assist in informed decision-making and contribute to the improvement of agricultural productivity and planning.

1.2 Purpose

The purpose of this project is:

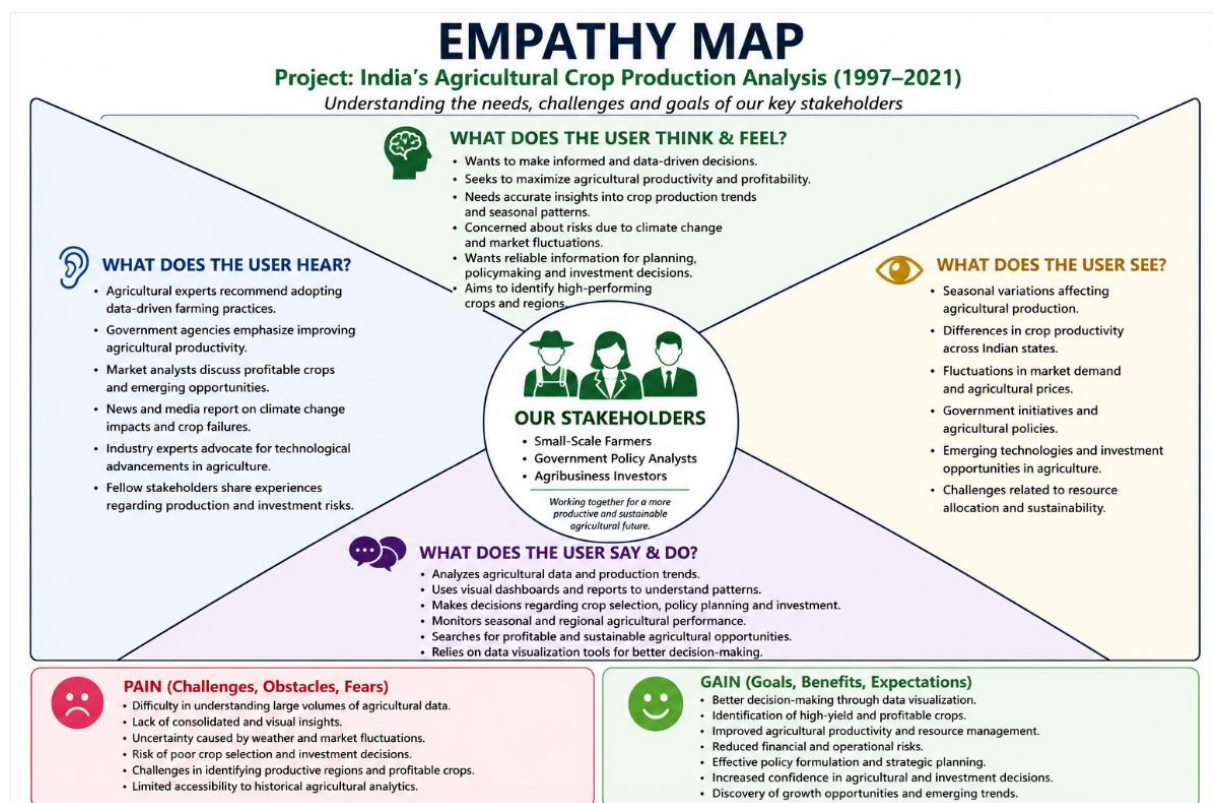
- To analyze crop production trends in India.
- To identify high-performing states and crops.
- To study seasonal agricultural patterns.
- To develop interactive Tableau dashboards.
- To assist stakeholders in making data-driven decisions.

CHAPTER-2 : IDEATION PHASE

2.1 Problem Statement

Farmers, agricultural policymakers, and researchers struggle to analyze large agricultural datasets due to fragmented information and lack of visual insights. This project aims to provide a comprehensive analytical dashboard that enables effective decision-making through interactive visualizations.

2.2 Empathy Map Canvas



2.3 Brainstorming

Team members brainstormed ideas and identified the major challenges:

- State-wise production analysis
- Crop performance analysis
- Seasonal trend analysis
- Yield comparison
- Dashboard storytelling

Team Members:

Member	Responsibility
Garima Rao	Data Collection
Mohan Singh Shekhawat	Visualization Design
Yusuf Khan	Dashboard Development
Komal Goyal	Story Creation & Documentation
Durlabh Kaushal	Testing & Deployment

CHAPTER-3: REQUIREMENT ANALYSIS

3.1 Customer Journey Map



3.2 Solution Requirement

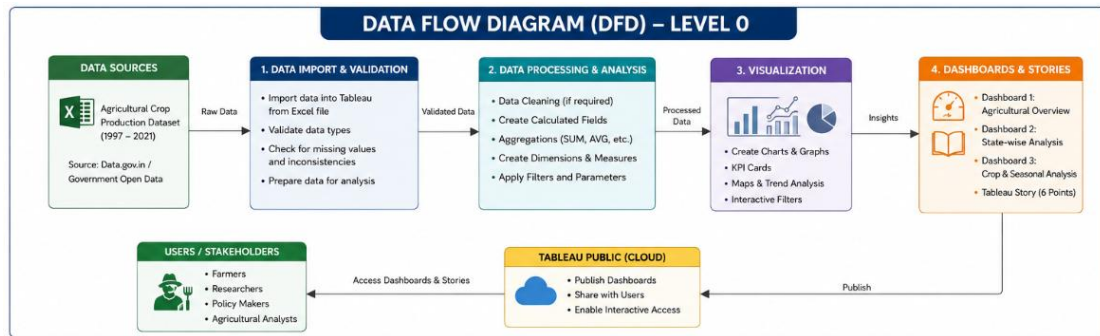
Functional Requirements are as follows:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Import & Processing	Import dataset, validate data, create calculated fields
FR-2	Dashboard Development	Agricultural Overview, State Analysis, Crop Analysis dashboards
FR-3	Interactive Visualization	Filters, charts, maps, KPI cards, drill-down analysis
FR-4	Story & Publishing	Tableau Story creation, insights generation, Tableau Public publishing

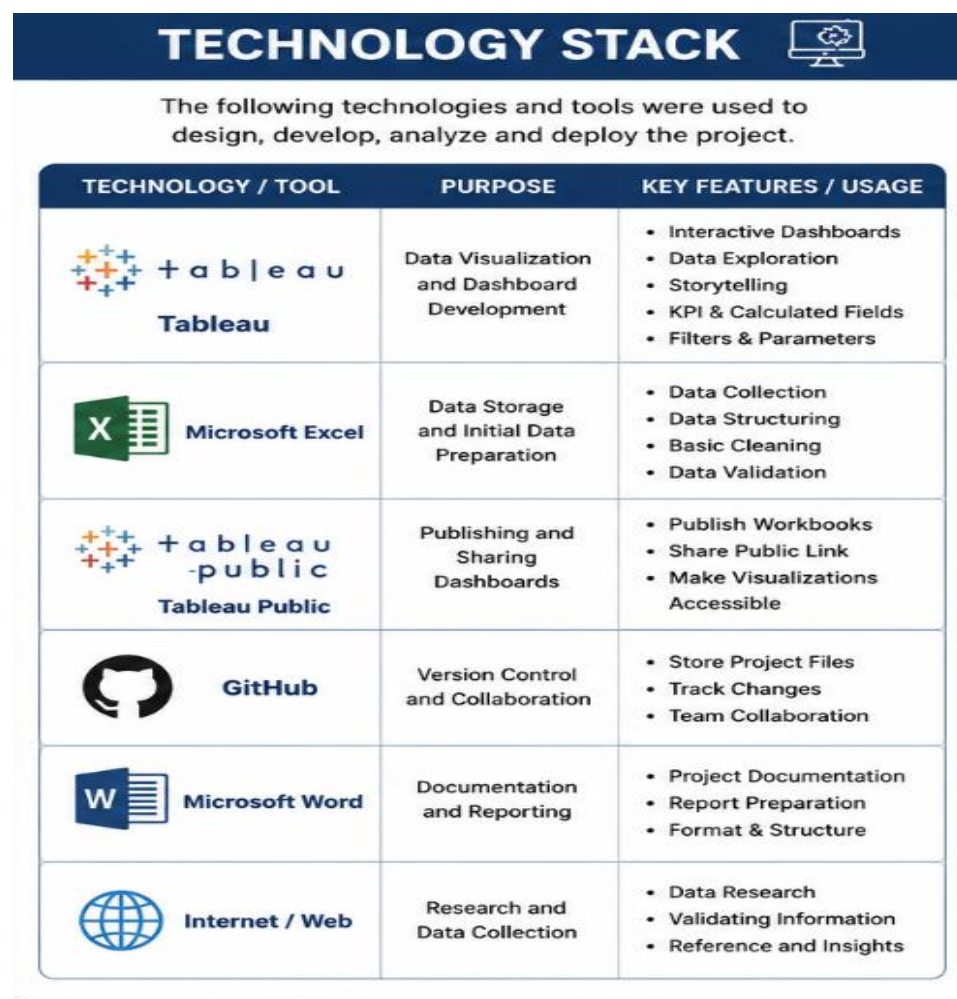
Non Functional Requirements are as follows:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboards should provide an intuitive and user-friendly interface with interactive filters, charts, and visualizations that can be easily understood by farmers, researchers, and policymakers.
NFR-2	Security	The project uses publicly available agricultural data and is published through Tableau Public, ensuring safe access without requiring user authentication or sensitive data handling.
NFR-3	Reliability	The system should consistently display accurate agricultural insights and maintain data integrity across all dashboards and story visualizations.
NFR-4	Performance	The dashboards should load efficiently and respond quickly to filter selections, drill-down operations, and visualization rendering.
NFR-5	Availability	The dashboards and stories published on Tableau Public should remain accessible to users anytime through an internet connection.
NFR-6	Scalability	The solution should support the addition of future agricultural datasets, new visualizations, and extended analysis without major redesign.

3.3 Data Flow Diagram



3.4 Technology Stack



CHAPTER-4 : PROJECT DESIGN

4.1 Problem Solution Fit

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS • Farmers • Agricultural Researchers • Policymakers / Government Agencies • Students and Data Analysts • Agricultural Organizations People who need insights on agricultural crop production trends to make informed decisions.	6. CUSTOMER CONSTRAINTS CC • Large volume of agricultural data • Data scattered across multiple reports and sources • Lack of interactive and easy-to-use analysis tools • Time-consuming manual analysis • Limited visualization and comparison capabilities Constraints that prevent users from taking action or limit their choices.	5. AVAILABLE SOLUTIONS AS • Government agricultural reports • Excel sheets and static charts • Agricultural surveys and publications • Research papers • Basic data tables on websites Solutions are available but are not interactive, hard to analyze or not user-friendly.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P • Analyze crop production trends across years (1997–2021) • Compare crop performance across states • Identify top and low producing crops • Understand seasonal production patterns • Support agricultural planning and policy decisions • Provide data-driven insights for research and studies	9. PROBLEM ROOT CAUSE RC • Agricultural data exists but is not centralized. • Lack of an interactive platform for deep analysis. • Difficult to derive meaningful insights from raw data. • No easy way to visualize trends, compare states, crops, seasons and years together. The real reason behind the problem. What is the root cause?	7. BEHAVIOUR BE • Users search for data in reports, spreadsheets and government portals • Compare numbers manually across different sources • Create their own charts in Excel • Spend more time cleaning and understanding data than analyzing it	
Focus on J&P, top into BE, understand RC	3. TRIGGERS TR • Need for agricultural planning • Policy formulation and review • Research and academic projects • Demand for state and crop comparison • Seasonal planning and forecasting	10. YOUR SOLUTION SL India Agricultural Crop Production Analysis (1997–2021) Using Tableau • Interactive dashboards with KPIs and trend analysis • State-wise, crop-wise, and seasonal comparison • Maps, charts and filters for easy exploration • Tableau Stories for data-driven narrative • Helps users make informed and data-driven decisions quickly A centralized, interactive and visual analytics solution for agricultural crop production analysis.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE • Government portals (data source) • Tableau dashboards (interactive analysis) • Research websites and academic resources 8.2 OFFLINE • Agricultural departments • Research institutions and universities • Workshops, seminars and training programs	Focus on J&P, top into BE, understand R
Identify Group TR & EM	4. EMOTIONS: BEFORE / AFTER EM Before: Confusion, Lack of insights, Overwhelmed by data, Uncertainty After: Clarity, Confidence, Better decision-making, Satisfaction, Data-driven planning			Extract online & offline CH of BE

4.2 Proposed Solution

The proposed system analyzes agricultural data through:

- Dashboard 1: Overview Analysis
- Dashboard 2: State Analysis
- Dashboard 3: Crop & Seasonal Analysis
- Story Points

4.3 Solution Architecture

Agricultural Dataset



Data Validation



Tableau



KPI Calculation



Visualizations



Dashboards



Story Analysis



Decision Support

CHAPTER-5 : PROJECT PLANNING & SCHEDULING

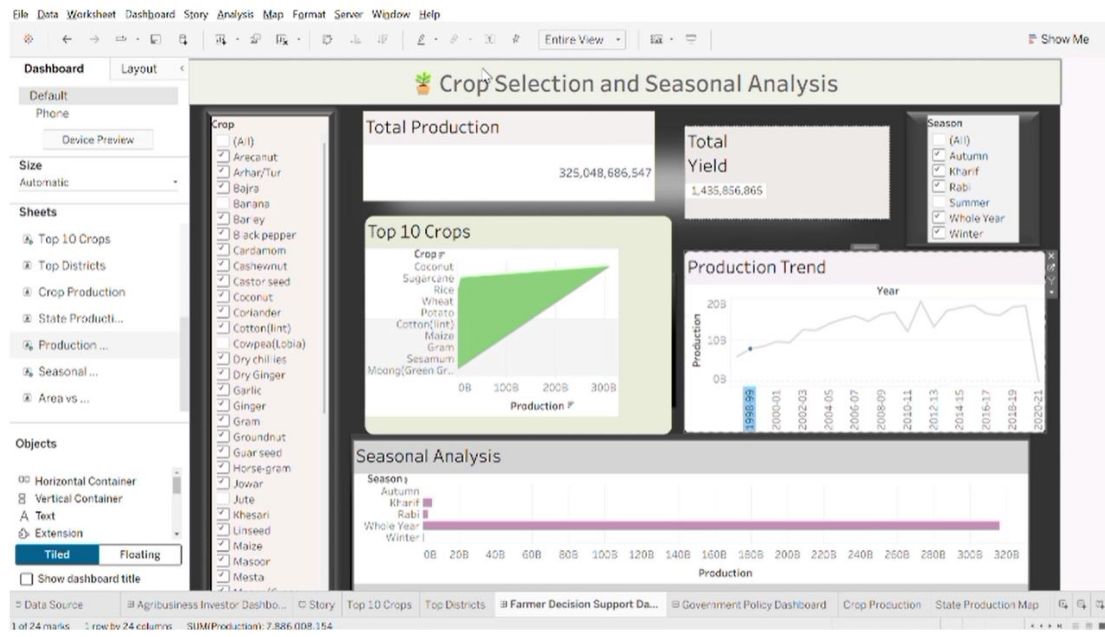
Activity	Duration
Project Identification	1 day
Data Collection	1 day
Dashboard Design	1 day
Story Development	1 day
Testing	1 day
Deployment	1 day

CHAPTER-6 : FUNCTIONAL & PERFORMANCE TESTING

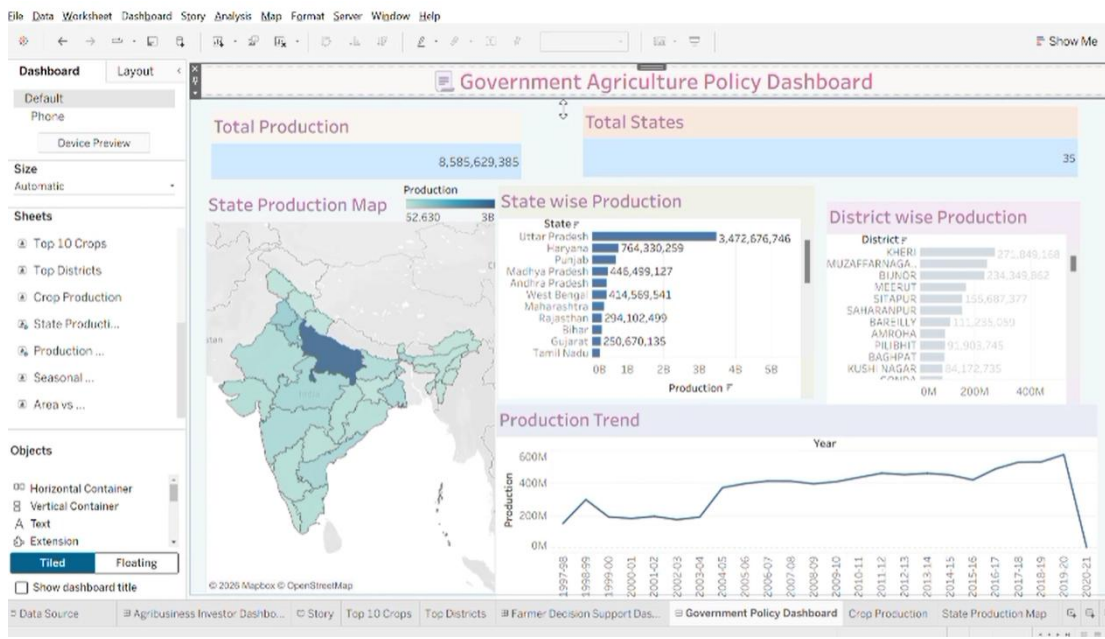
Parameter	Result
Data Rendering	Successful
Data Validation	Successful
Filters	Working
Calculated Fields	Working
Dashboard Design	3 Dashboards
Story Design	6 Story Points

CHAPTER-7 : RESULTS

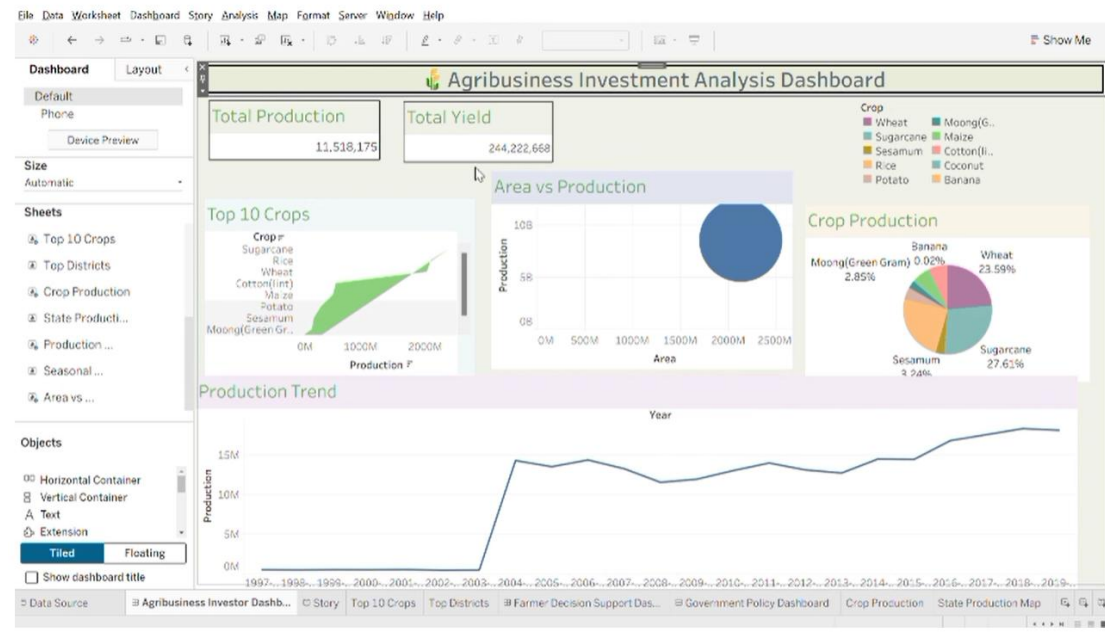
Dashboard 1



Dashboard 2



Dashboard 3



CHAPTER-8 : ADVANTAGES & DISADVANTAGES

Advantages

- Interactive dashboards
- Easy analysis
- Better decision-making
- User-friendly visualization
- Dynamic filtering

Disadvantages

- Depends on dataset quality
- Requires Tableau knowledge
- Limited predictive capability

CHAPTER-9 : CONCLUSION

This project successfully analyzed India's agricultural crop production from 1997 to 2021 using Tableau. Interactive dashboards and story visualizations provided meaningful insights regarding crop production trends, seasonal patterns, and state-wise agricultural performance.

CHAPTER-10 : FUTURE SCOPE

Future enhancements include:

- Machine Learning prediction
- Real-time agricultural data integration
- Weather forecasting analysis
- AI-powered recommendations
- Mobile dashboard development

CHAPTER 11 : APPENDIX

- [Tableau Workbook\(Dataset\)](#)
- [Tableau Public Link](#)
- [Github Repository Link](#)
- [Project Demo](#)